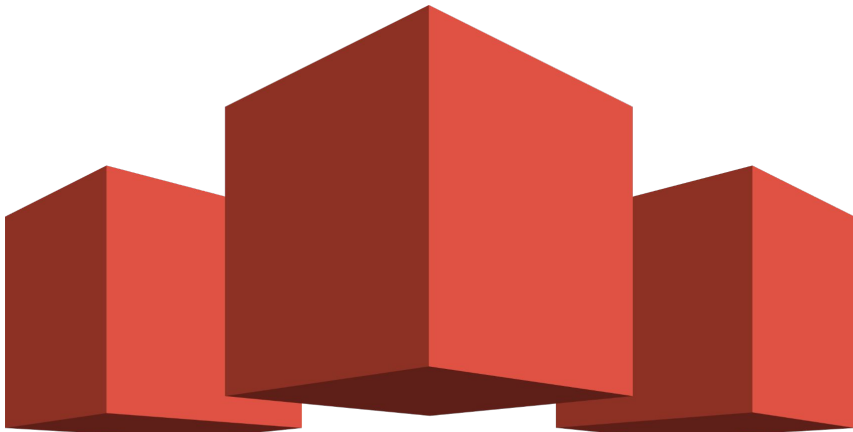


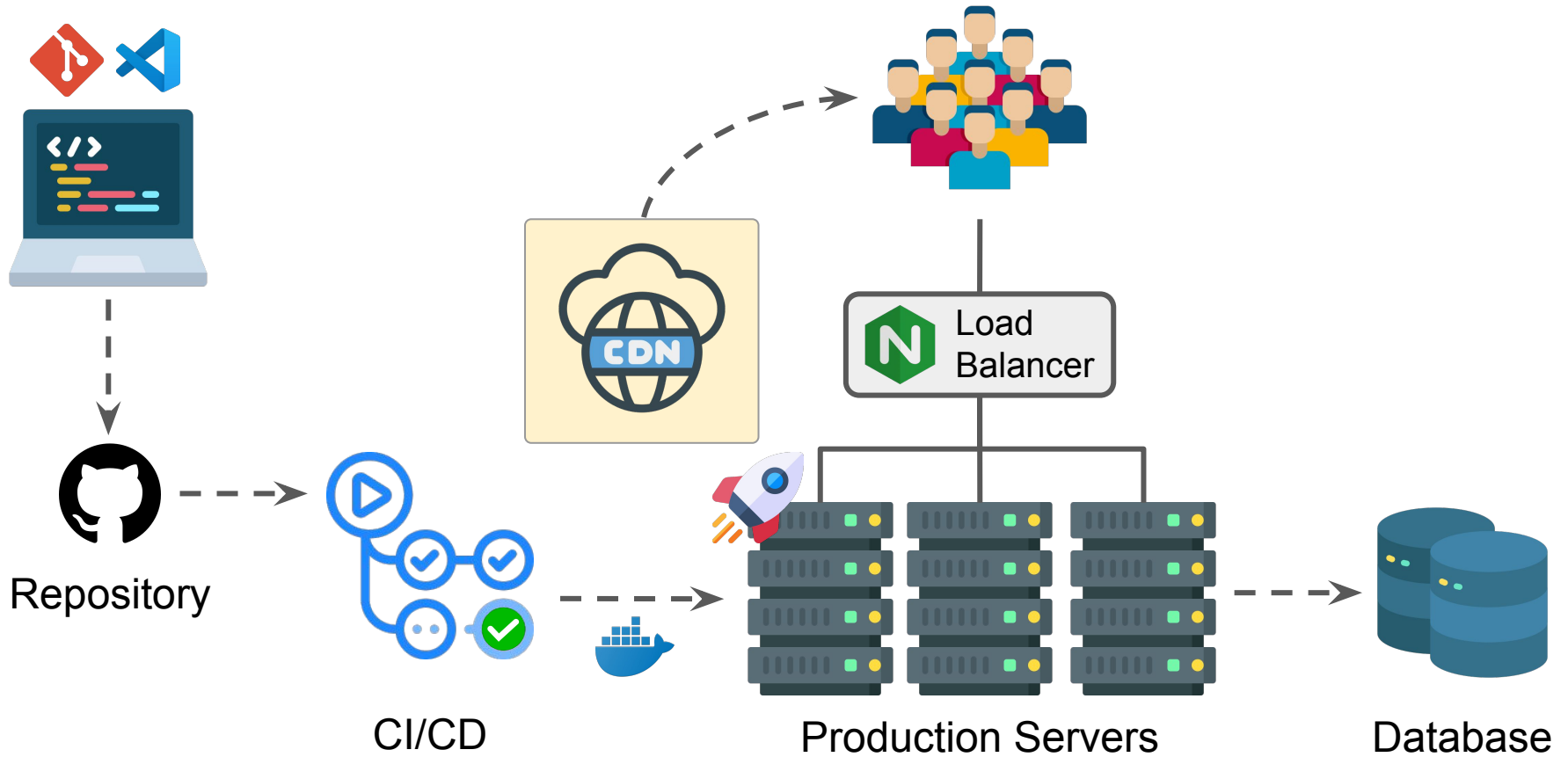


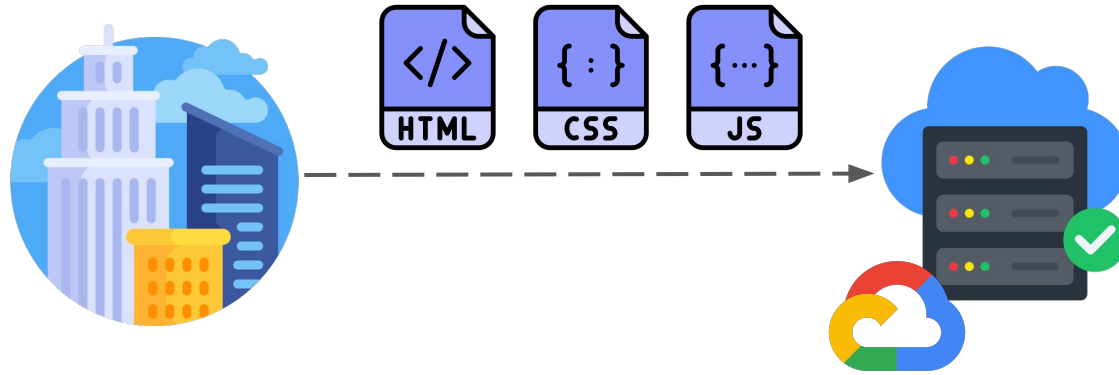
Caching and CDN



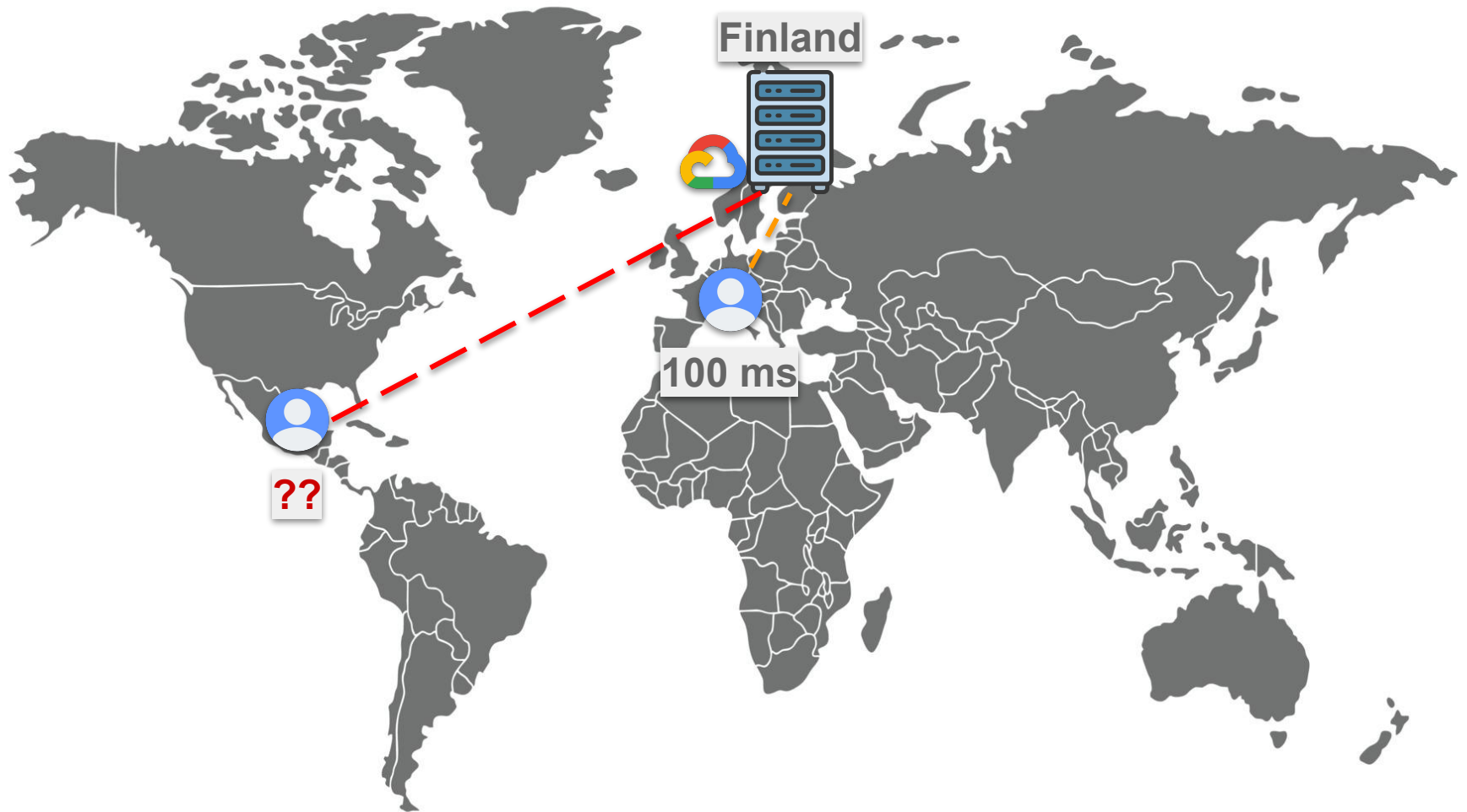
Mahir Labib Dihan
Lecturer, CSE, BUET

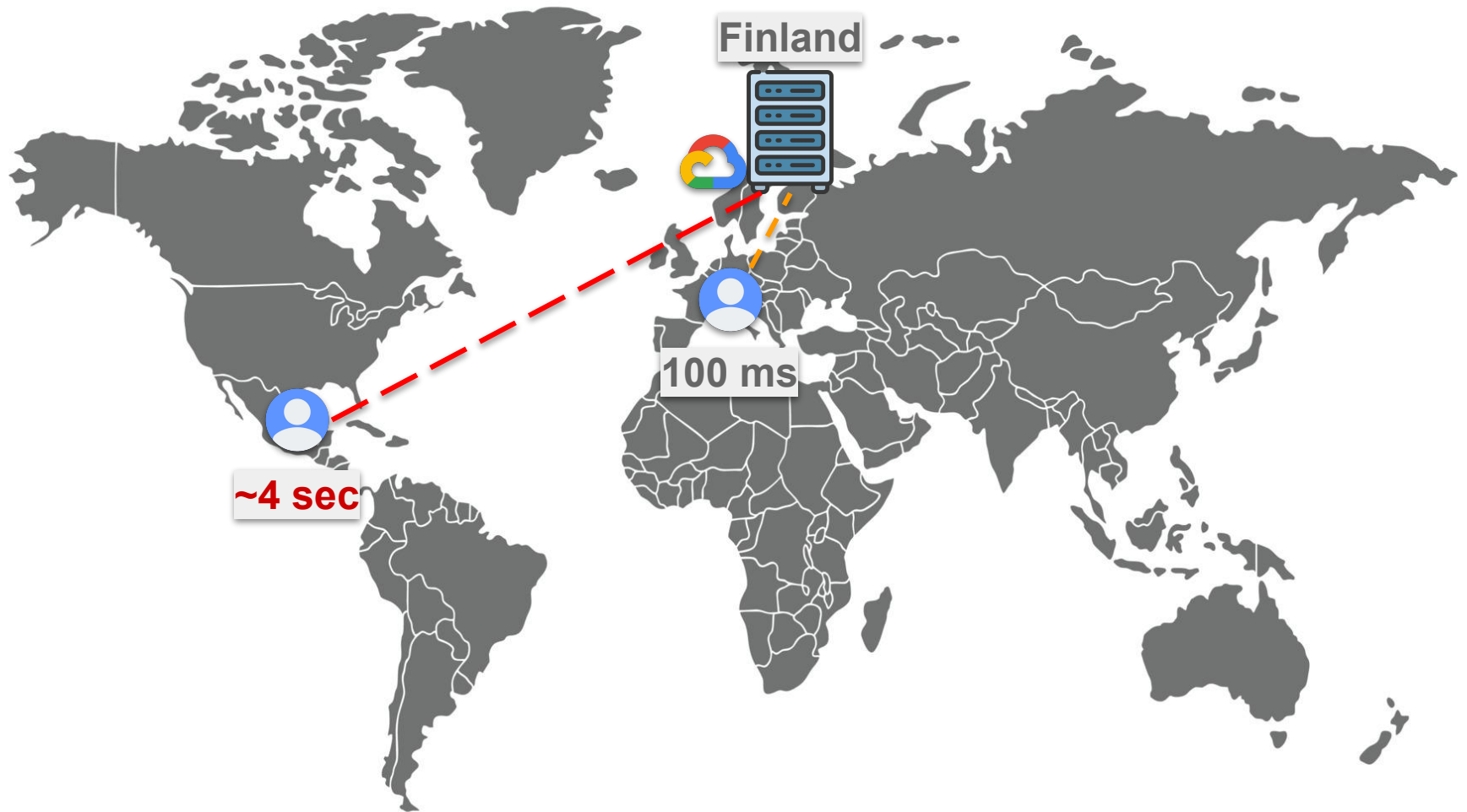


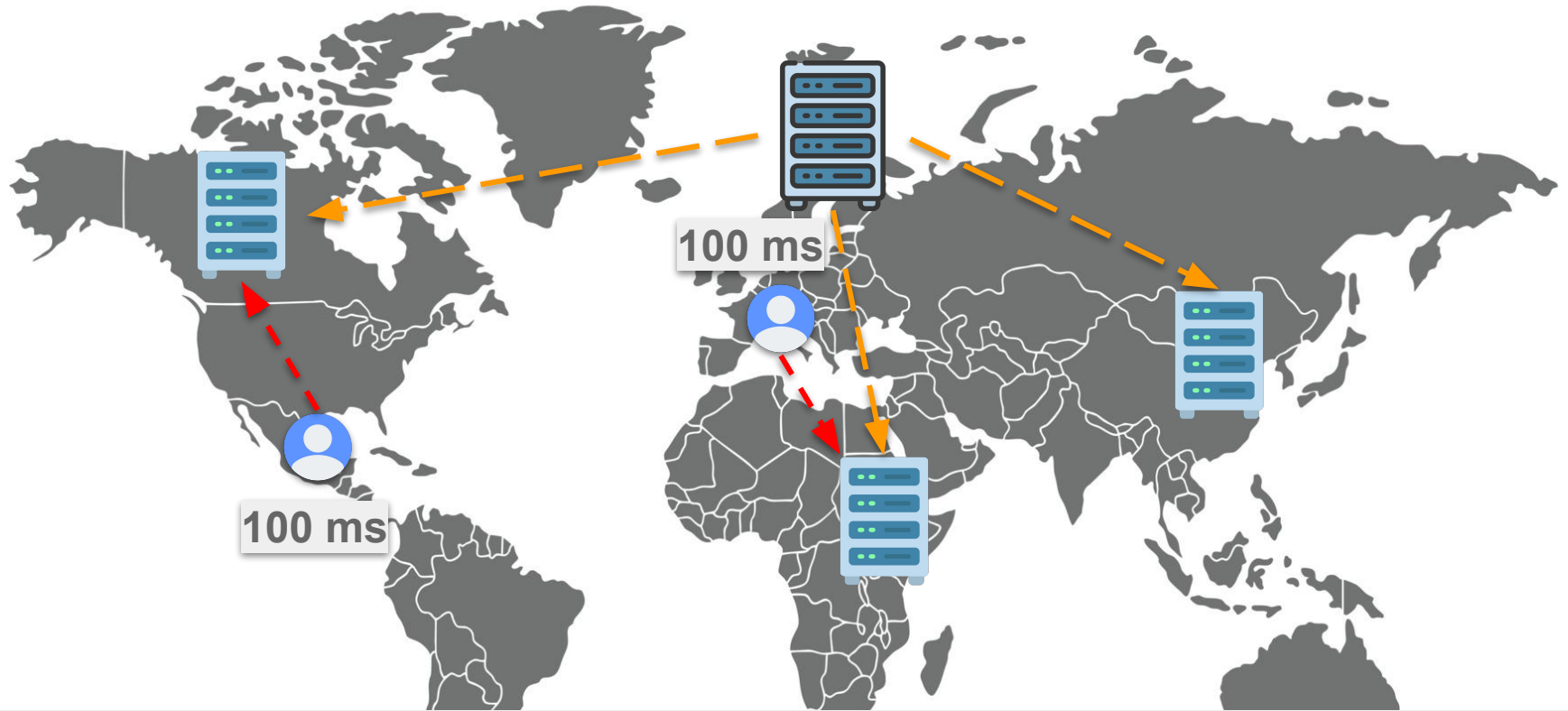




Let's assume a Company is hosting a website in Google Cloud Datacenter

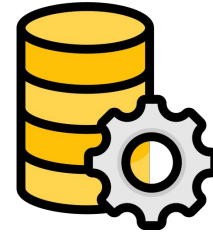






There are strategies to minimize the latency for users who are far away.

Caching

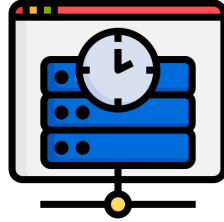


CDN

Content Delivery Network

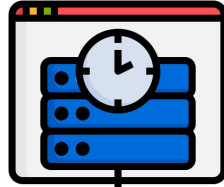


Caching

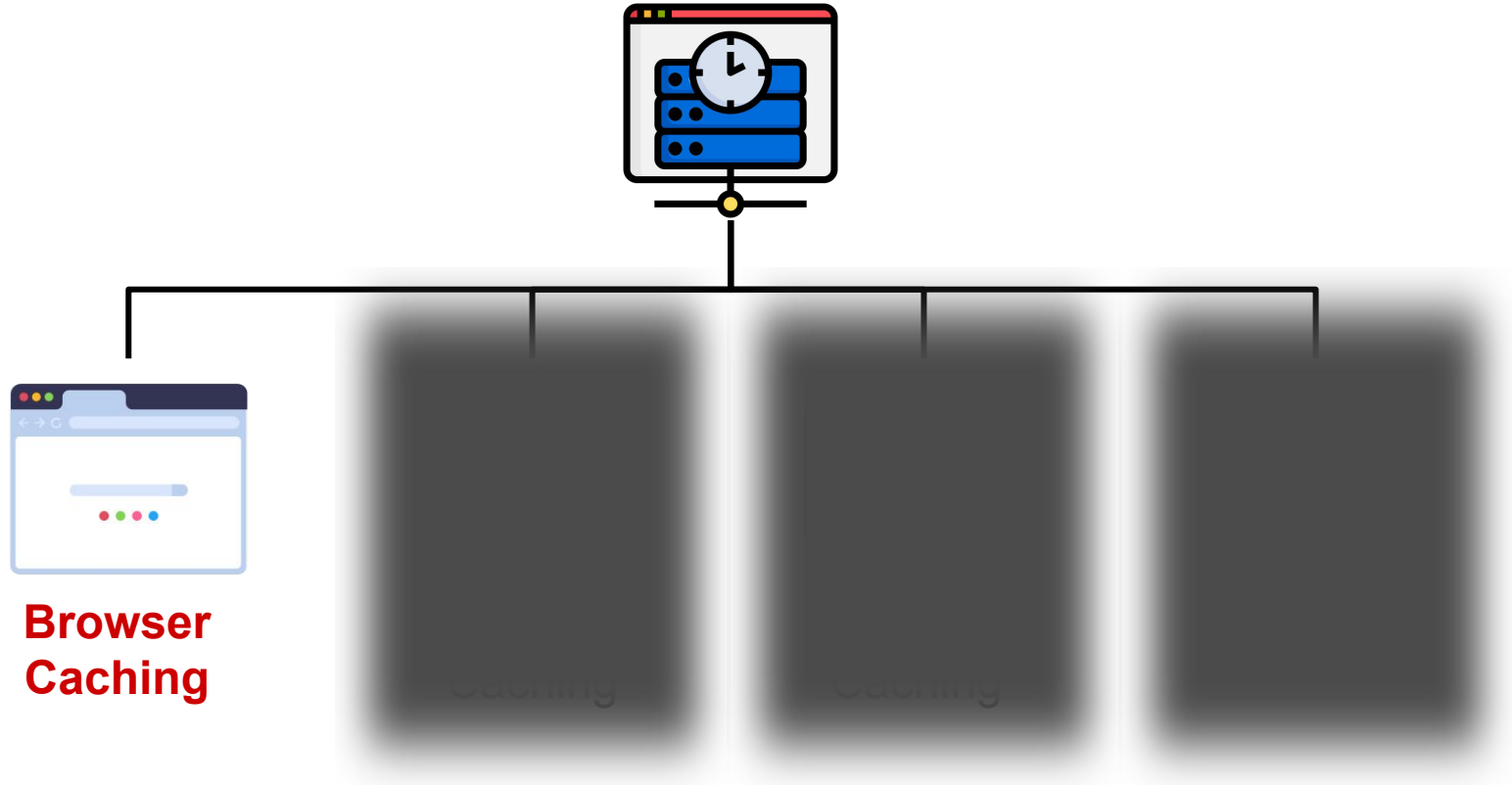


Storing a copy of data in a temporary storage (cache)

Caching

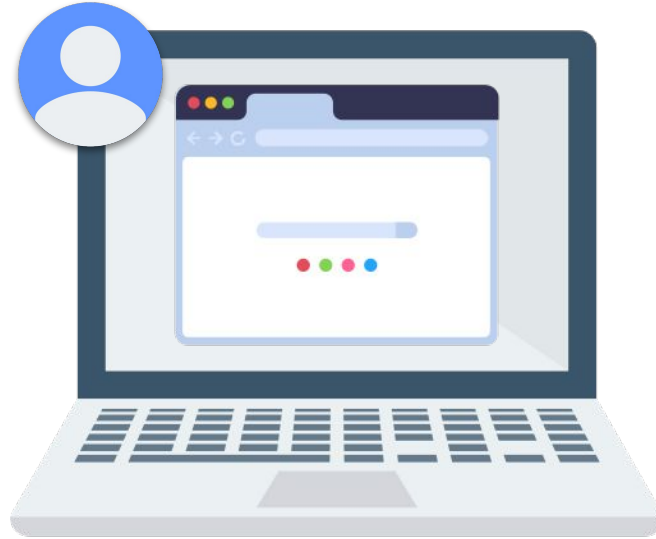


Caching



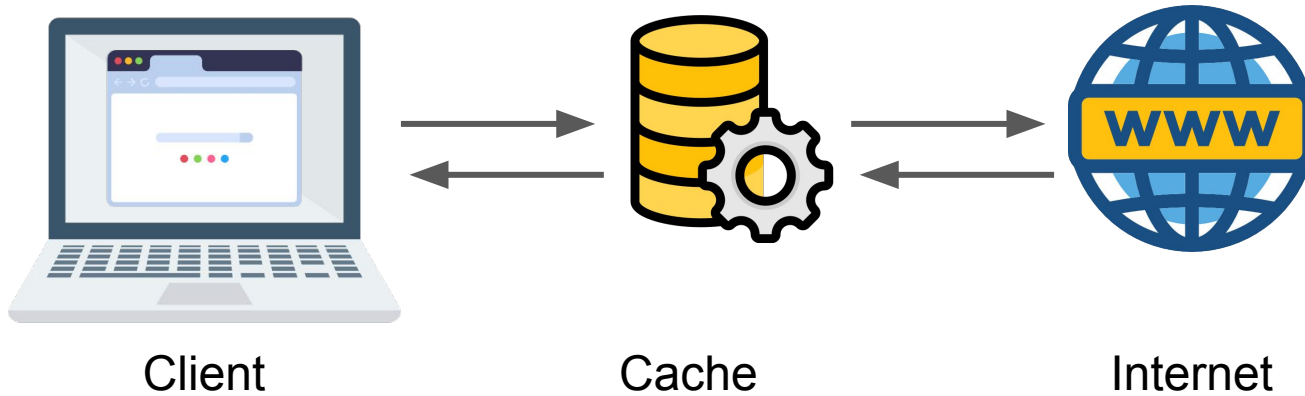
Browser Caching

Storage Location: Client's Computer

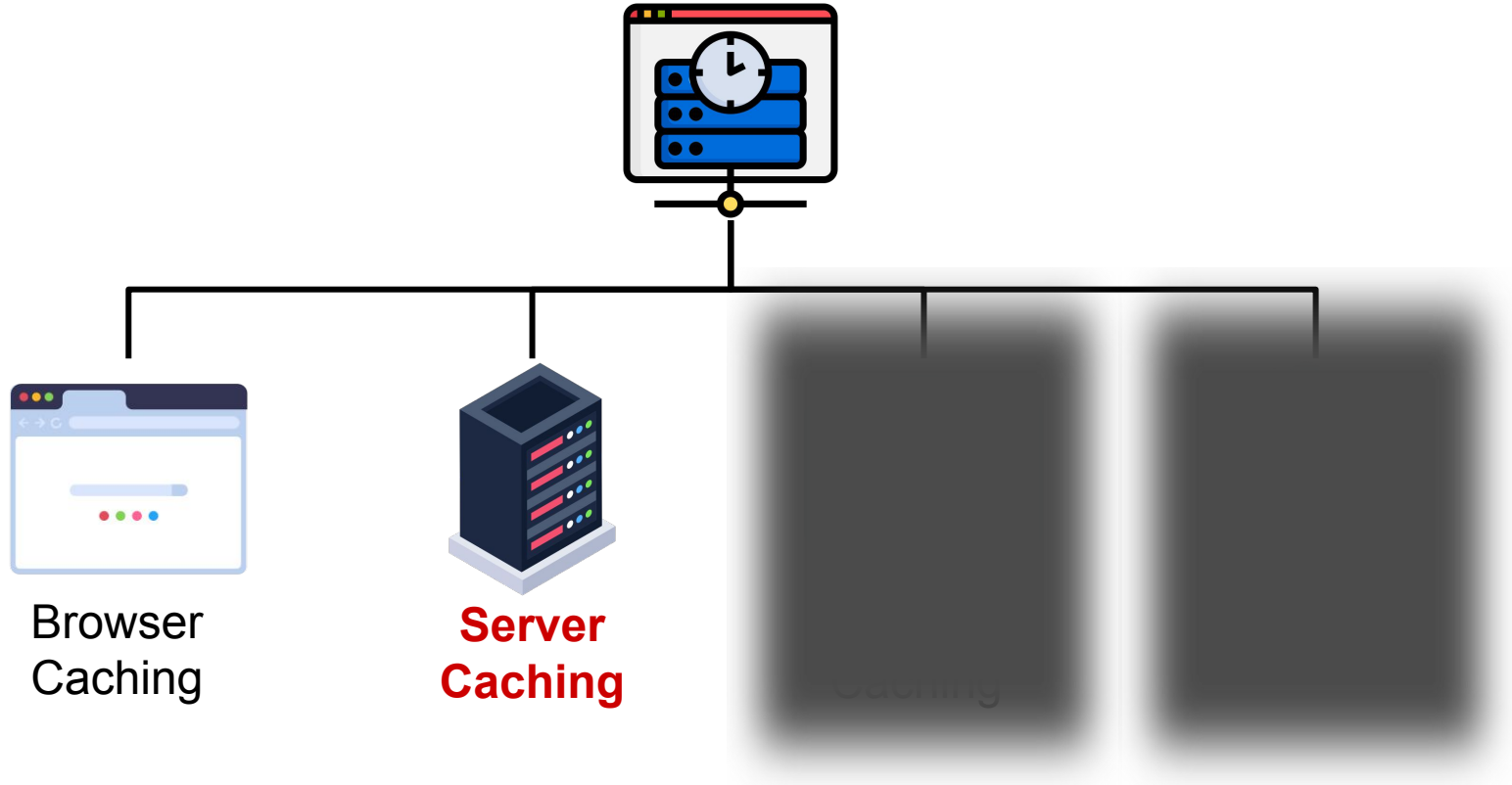


Browser Caching

Storage Location: Client's Computer



Caching



Server Caching

Storage Location: Inside or close to server

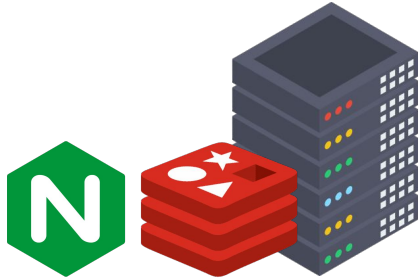
Involves storing frequently accessed data on the server side, reducing the need to perform expensive operations like database queries.

Server Caching

What is cached??

- API responses
- Computed results
- Session data
- Frequently requested objects (e.g., user profile, product list)

Server Caching

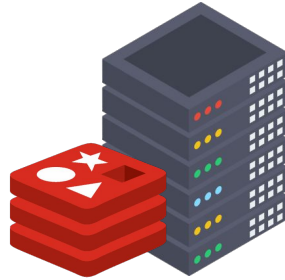


Server

Server Caching

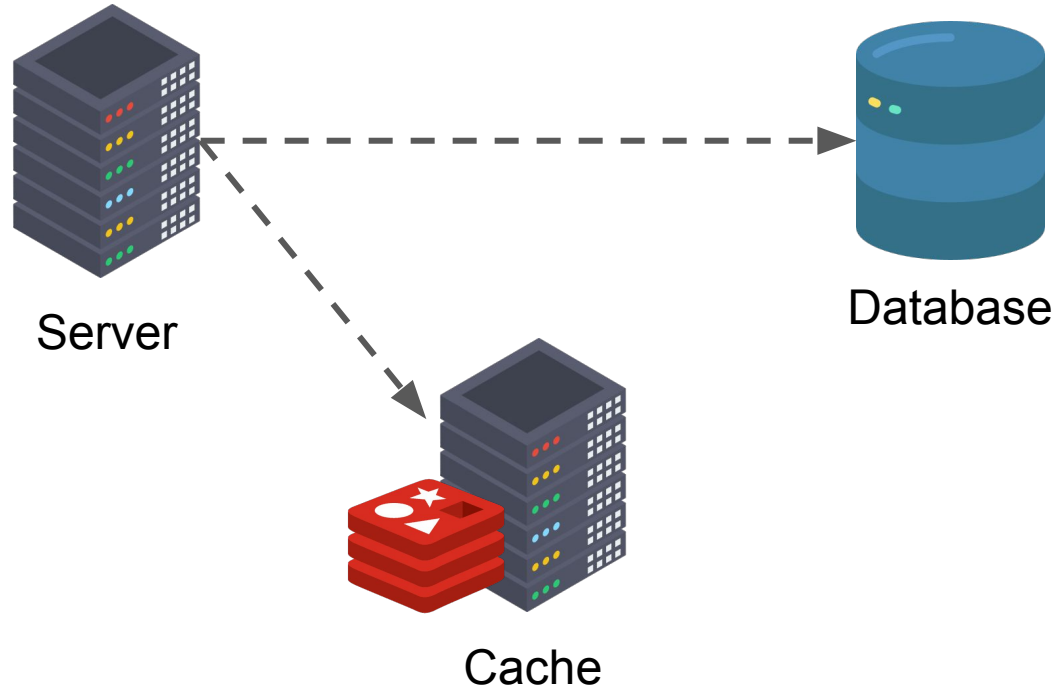


Server

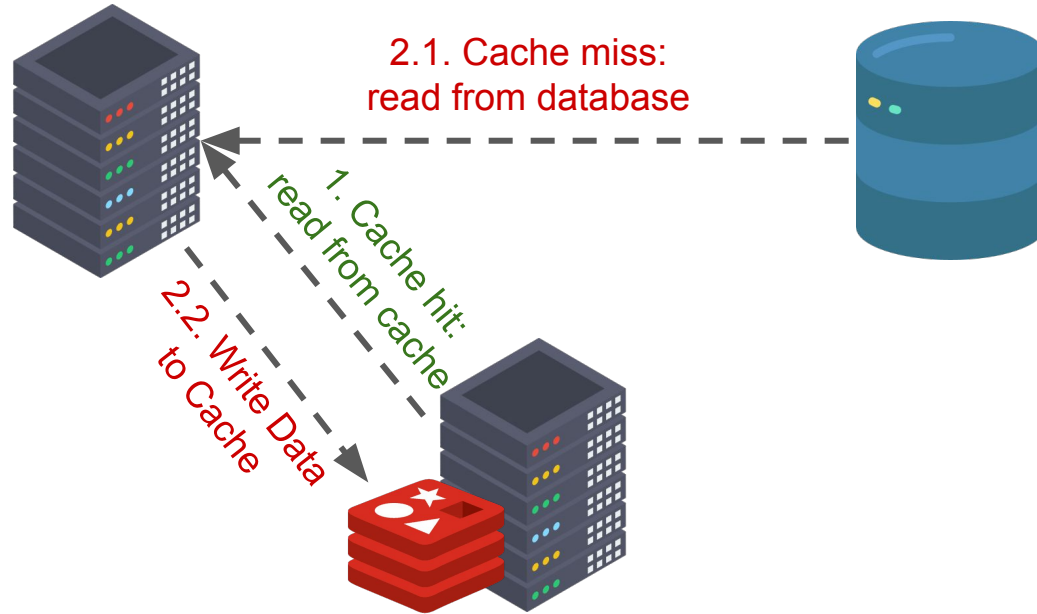


Cache

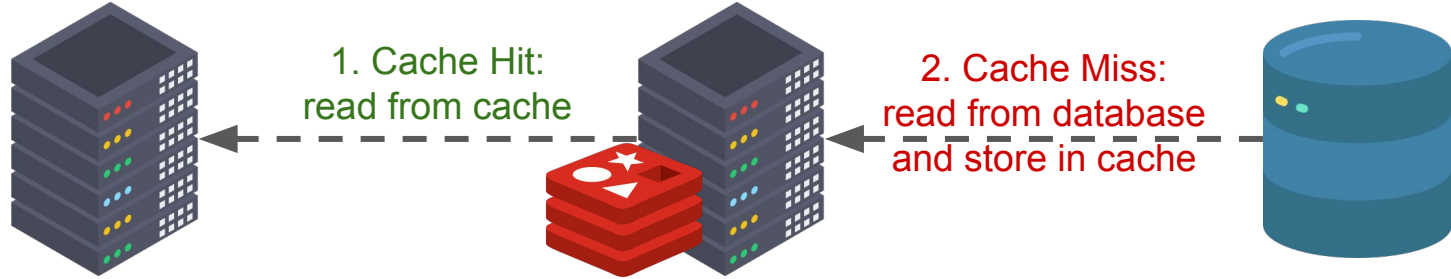
Server Caching



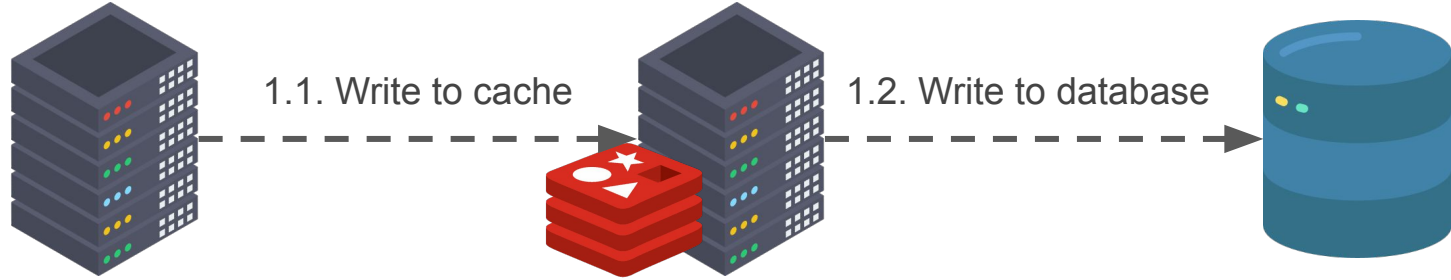
Cache-Aside



Read-Through

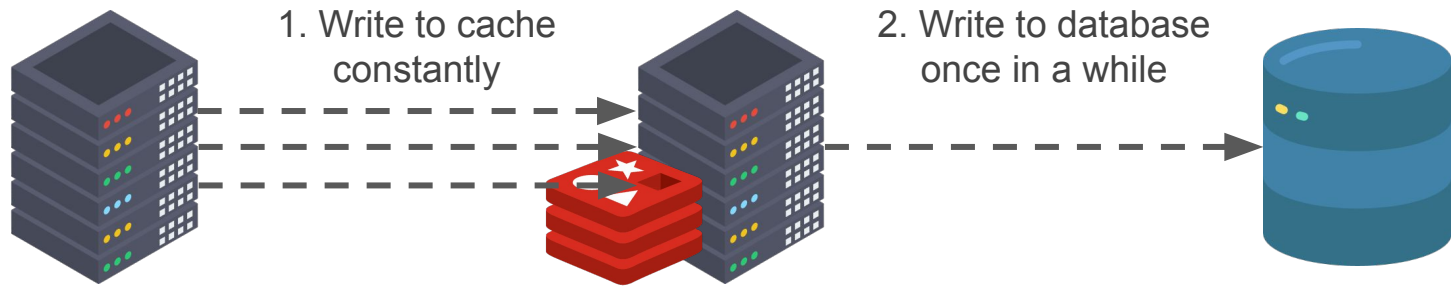


Write-Through



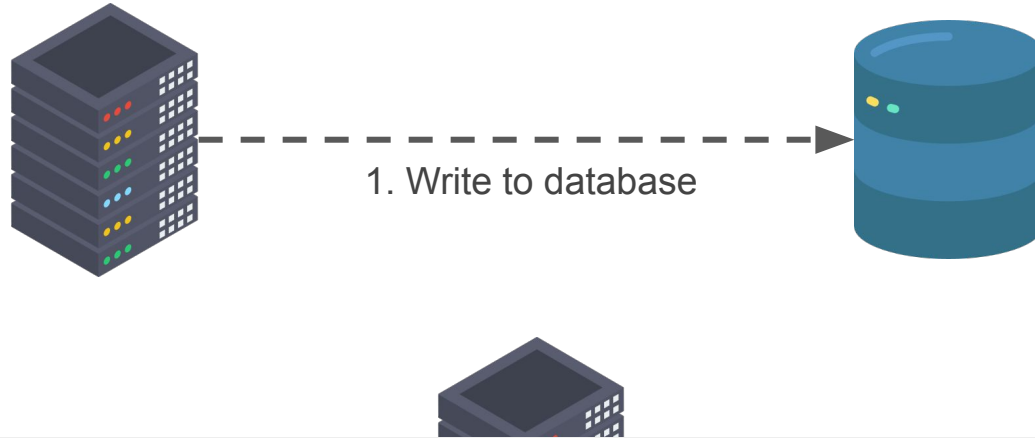
Ensures data consistency, but write can be slower.

Write-Back



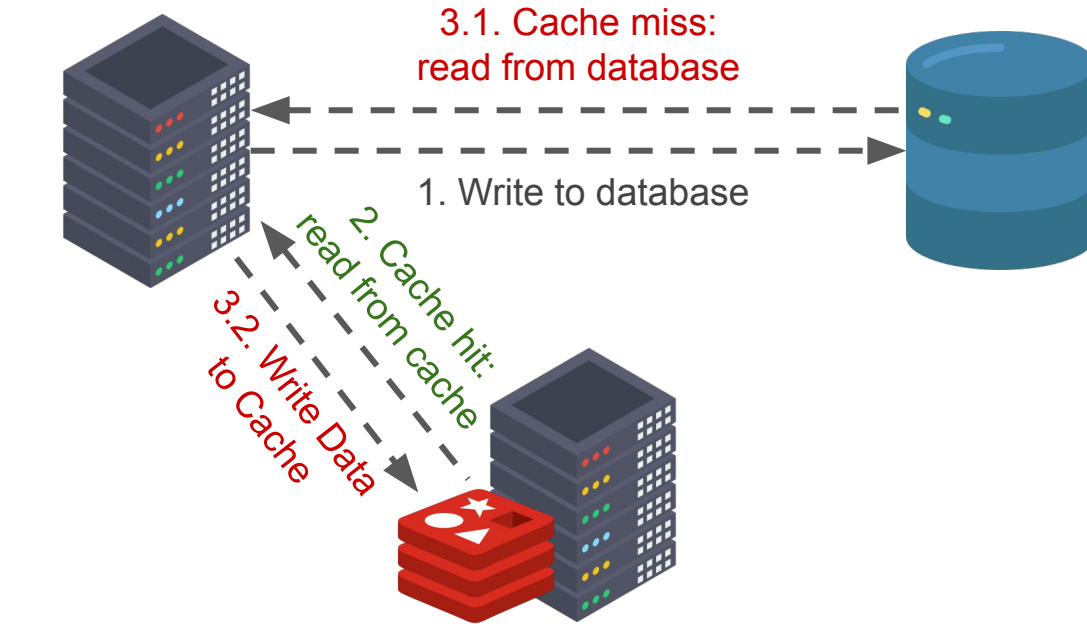
Improves write performance with a risk of losing data in case cache server crashes.

Write-Around

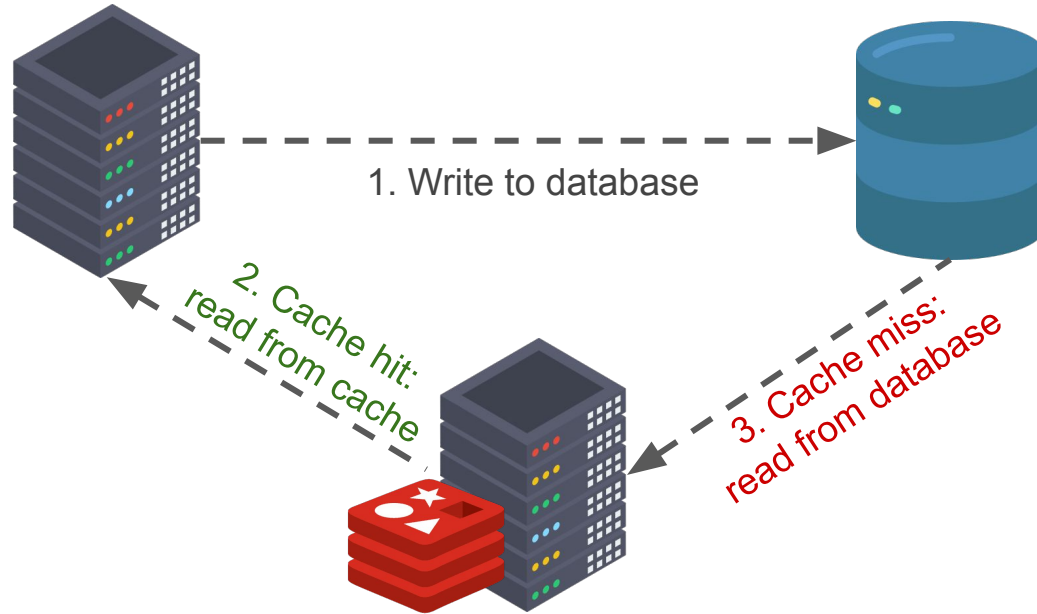


Data is written directly to the database and only the data that is read makes it way into the cache.

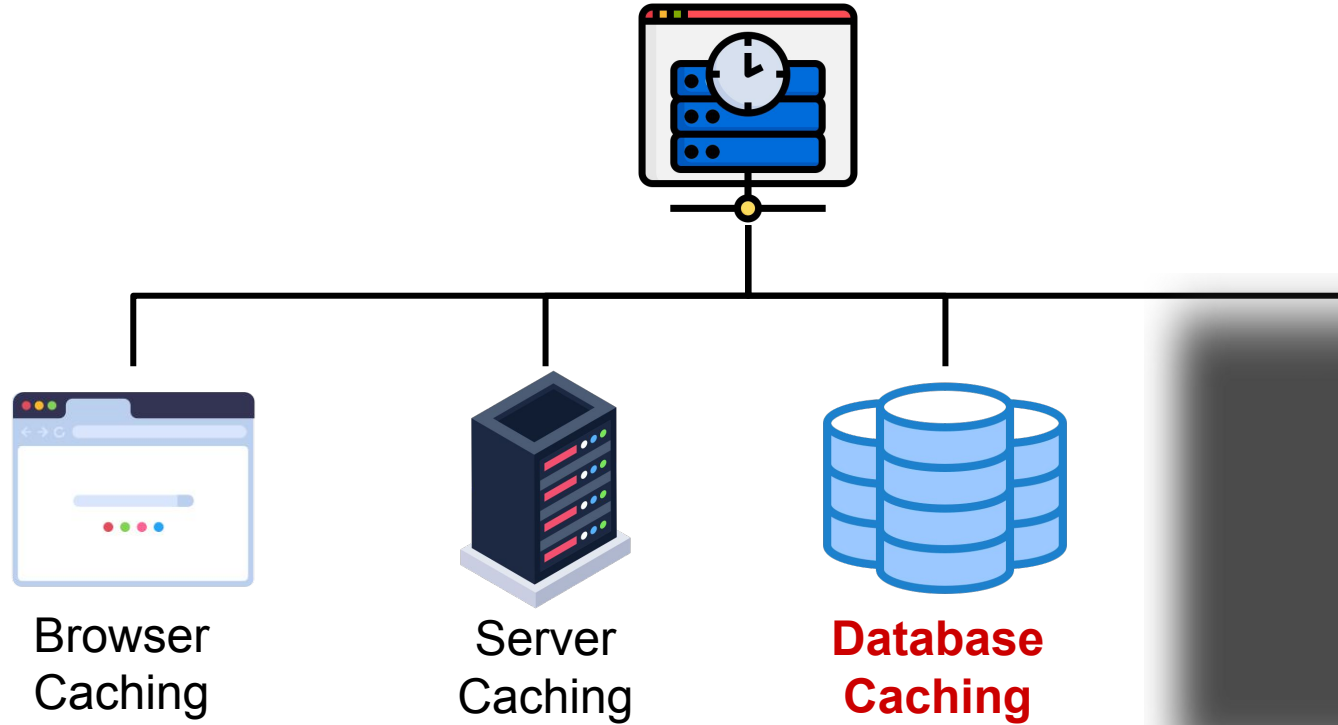
Write-Around + Cache Aside



Write-Around + Read-Through



Caching

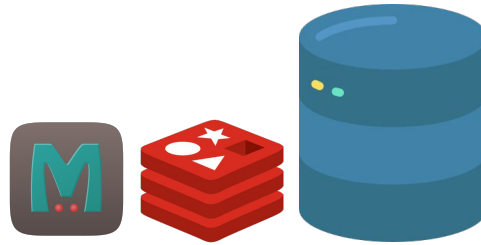


Database Caching

Storage Location: Inside or close to database

Practice of caching database query results to improve the performance of database-driven applications.

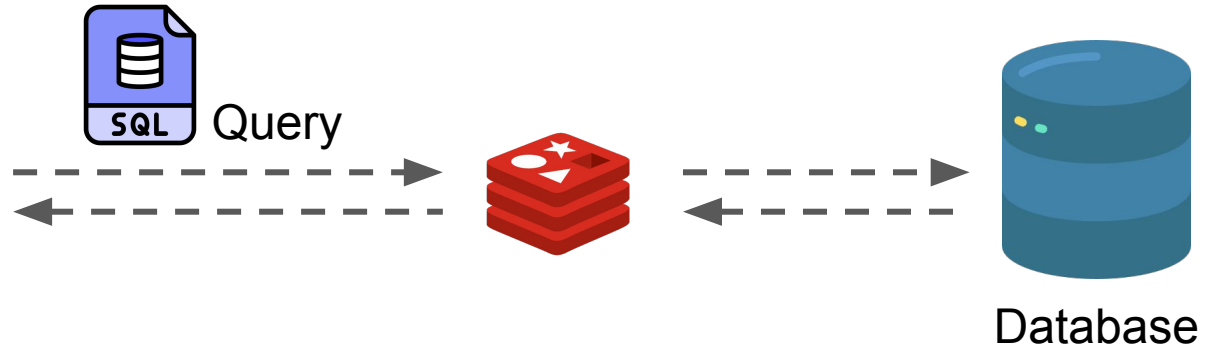
Database Caching



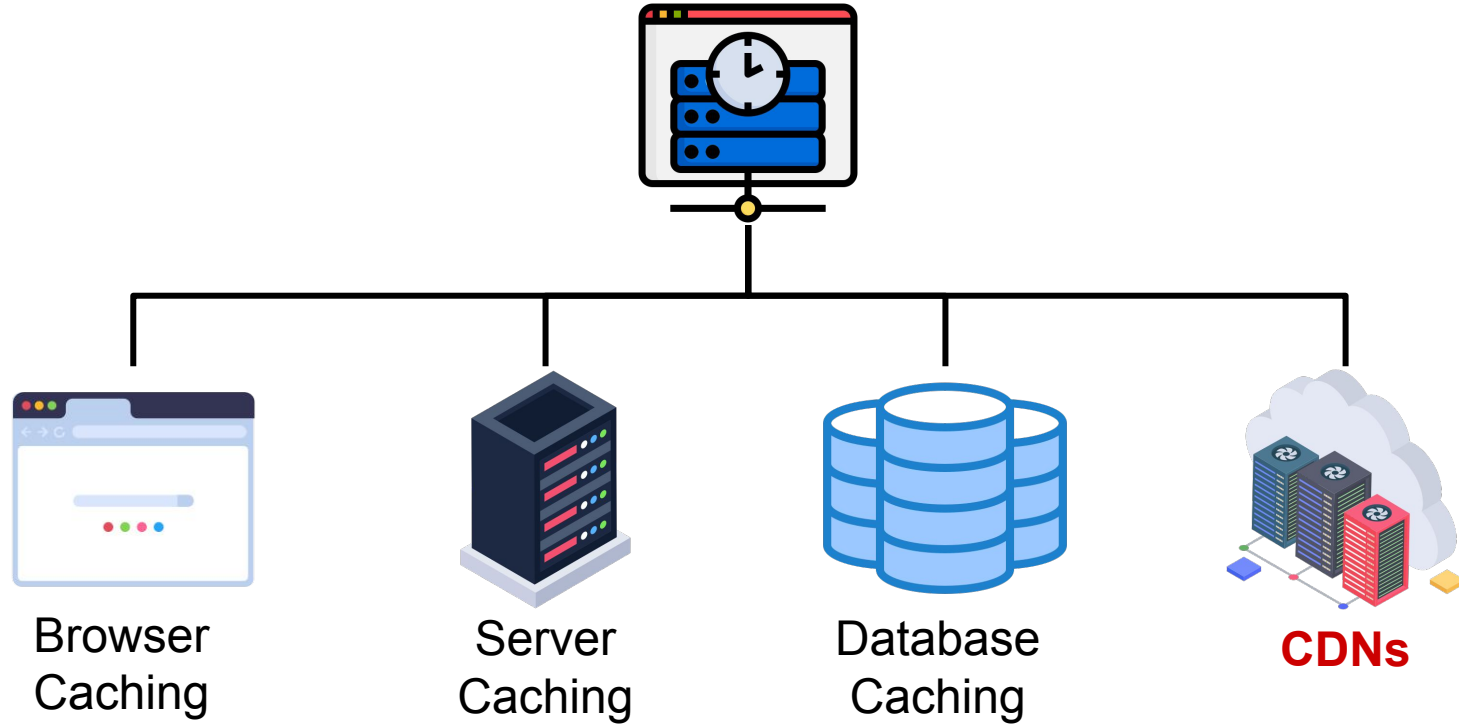
Database

Caching is often done either within the database system itself or via an external caching layer like redis or memcache.

Database Caching

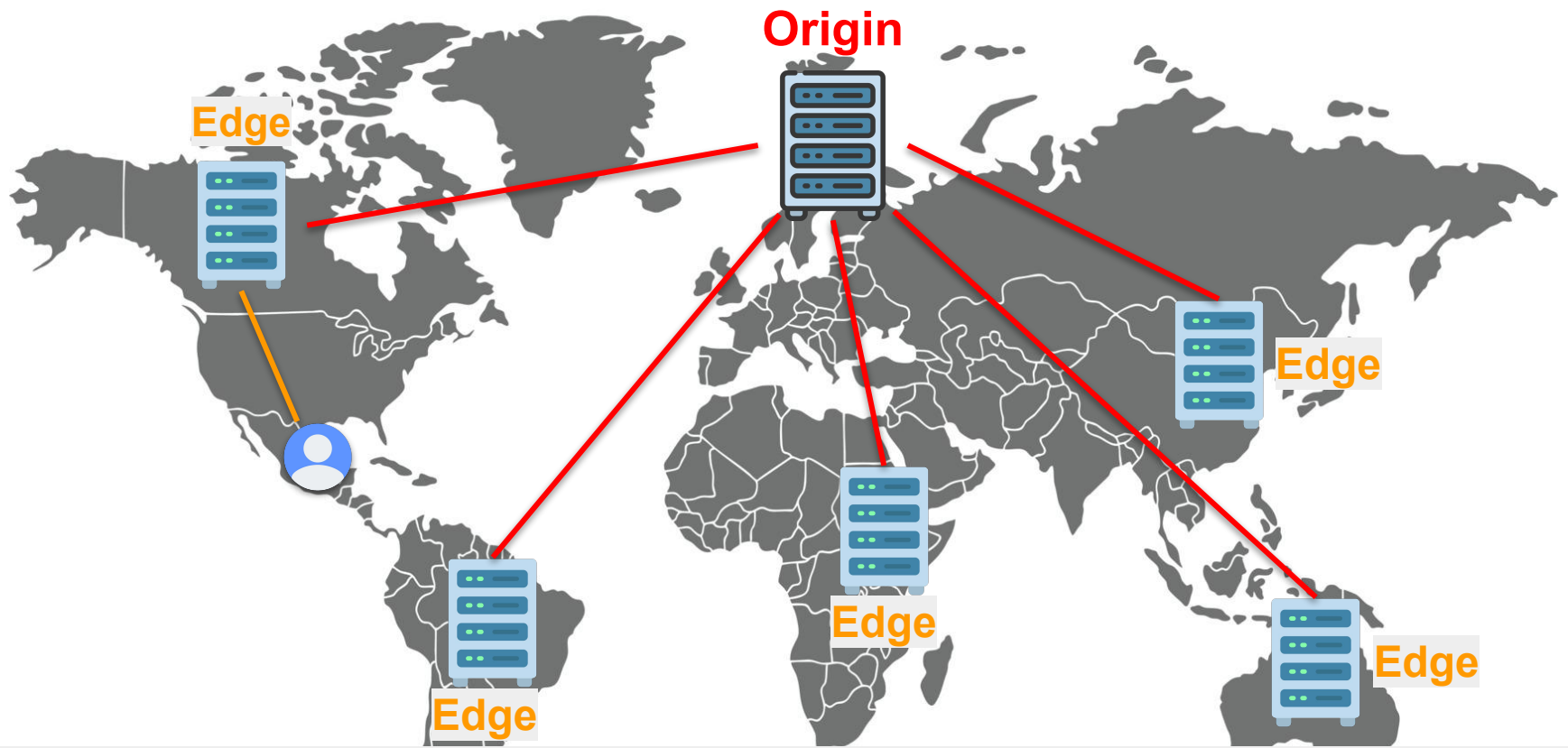


Caching





A network of servers which are distributed geographically. Generally used to serve static contents like HTML, CSS, JS or image and video files.



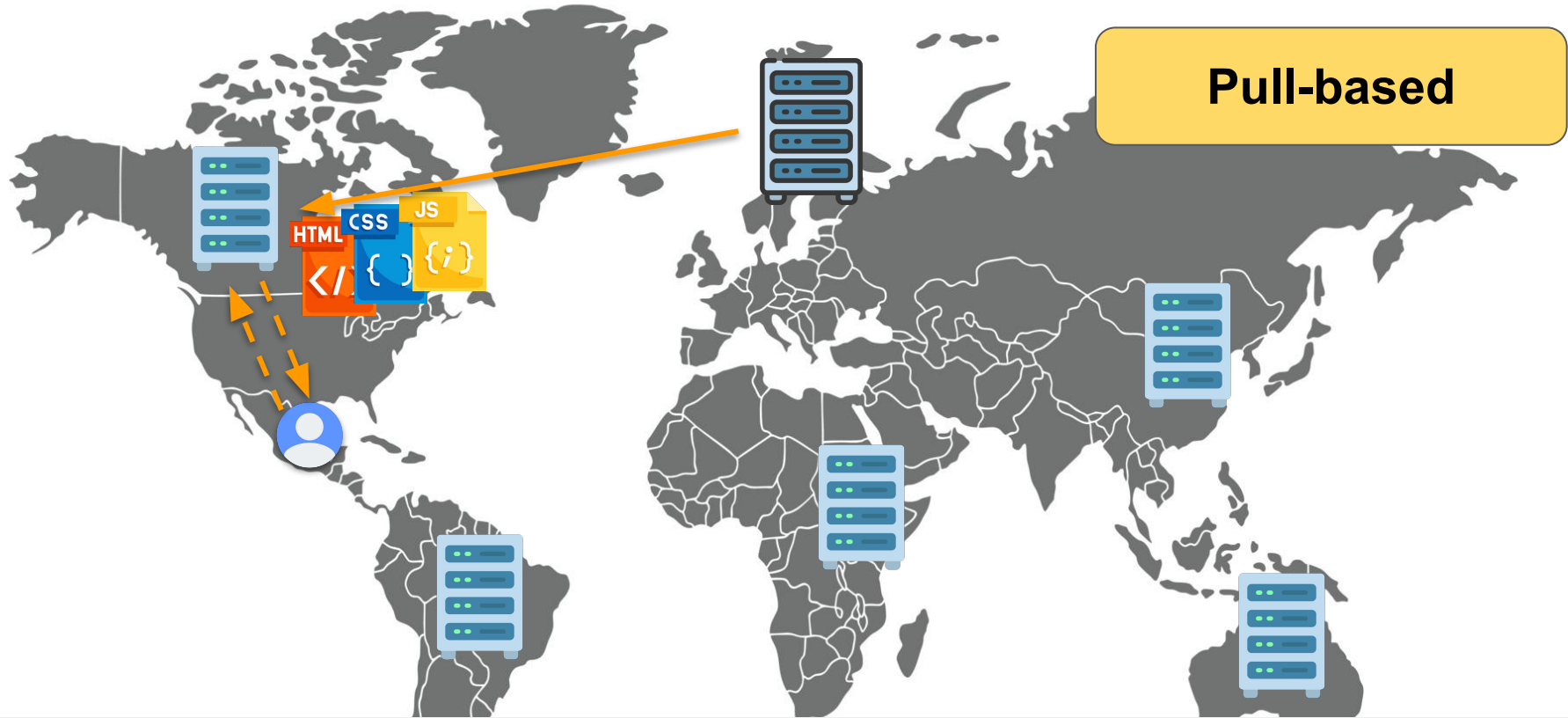
They cache the content from origin server and deliver it to users from the nearest CDN server.



When a user request for a file, the request is redirected to the nearest CDN server.



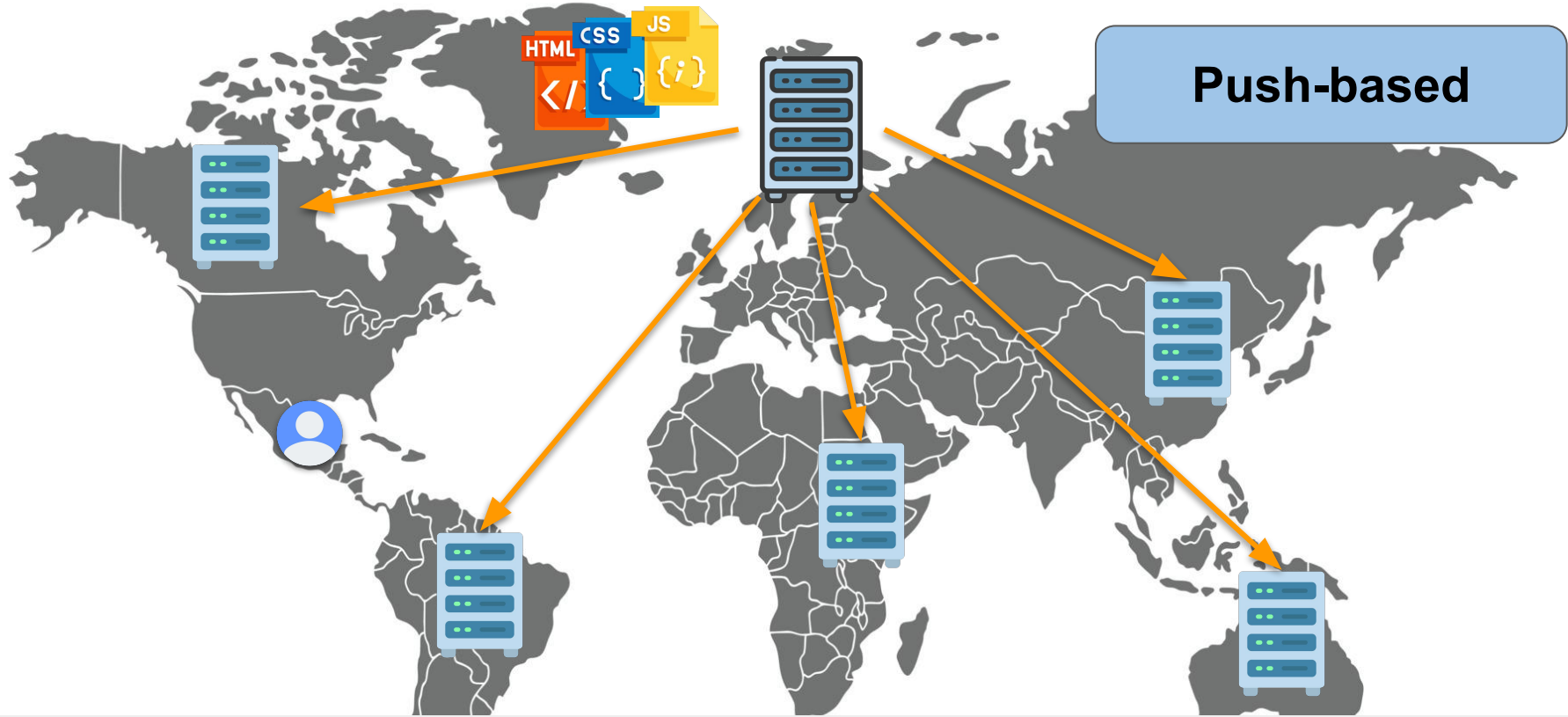
If the CDN server has the cached content, it delivers it to the user.



If not, it fetches the content from origin server, caches it and forwards to the user.

Pull-based CDN

- The CDN automatically pulls the content from Origin server when it's first requested by a user.
- It's ideal for websites with a lot of static content that is updated regularly.
- It requires less active management, because the CDN automatically keeps the content up-to-date.



You upload the content to the origin server, and then it distributes this files to the CDNs.

Push-based CDN

- This is useful when you have large files that are *infrequently* updated, but need to be quickly distributed when updated.
- It requires more active management of what content is stored on the CDNs.

Pull-based

- Reactive
- Ideal for websites with a lot of static content that is updated regularly
- Requires less active management

Push-based

- Proactive
- Useful when you have large files that are infrequently updated
- Requires more active management

When to Use CDNs and When to Hit the Origin Server

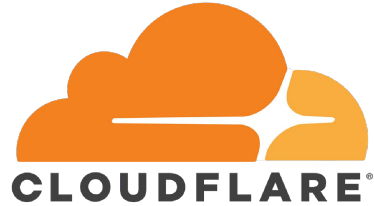
Use CDN When:

- Delivering static assets like images, CSS files, Javascript files and video content.
- You need to ensure high availability and performance for users across different geographical locations.
- Reducing the load on the origin server is a priority.

Use Origin Server When:

- Serving dynamic content that changes frequently or is personalized for individual users.
- Handling tasks that require real-time processing or access to up-to-date data.
- The application requires complex server-side logic that cannot be replicated or cached by a CDN.

CDN Provider



Pioneer of CDN (1998)

References

- https://www.youtube.com/watch?v=F2FmTdLtb_4&t=1759s